

```

1  #include <stdio.h>
2  #include <stdlib.h>
3  #include <string.h>
4  #include <unistd.h>
5  #include <sys/ipc.h>
6  #include <sys/shm.h>
7  #define SHM_SIZE 1024
8  int main() {
9      key_t key;
10     int shmid;
11     char *shm_ptr;
12     key = ftok("shmfile", 65);
13     shmid = shmget(key, SHM_SIZE, IPC_CREAT | 0666);
14     if (shmid == -1) {
15         perror("shmget");
16         exit(1);
17     }
18     shm_ptr = (char*) shmat(shmid, NULL, 0);
19     if (shm_ptr == (char*) -1) {
20         perror("shmat");
21         exit(1);
22     }
23     if (fork() == 0) {
24         sleep(1); // wait for parent to write
25         printf("Child reads: %s\n", shm_ptr);
26     } else {
27         strcpy(shm_ptr, "Hello from shared memory!");
28         printf("Parent wrote: %s\n", shm_ptr);
29     }
30     shmdt(shm_ptr);
31     shmctl(shmid, IPC_RMID, NULL);
32     return 0;
33 }

```

Data written to shared memory: Hello, shared memory!